

The PAR Master Defense Guide: Multi-Task ConvNeXt Architecture

> █

[SYSTEM CHECK] Leak-proof data pipeline ... VERIFIED
[SYSTEM CHECK] Mixed-precision engine ... VERIFIED
[SYSTEM CHECK] Multi-task hard-parameter sharing ... VERIFIED

Take a deep breath. You are walking in with a Principal-level pipeline. You built this. You own the logic. You are not cooked.

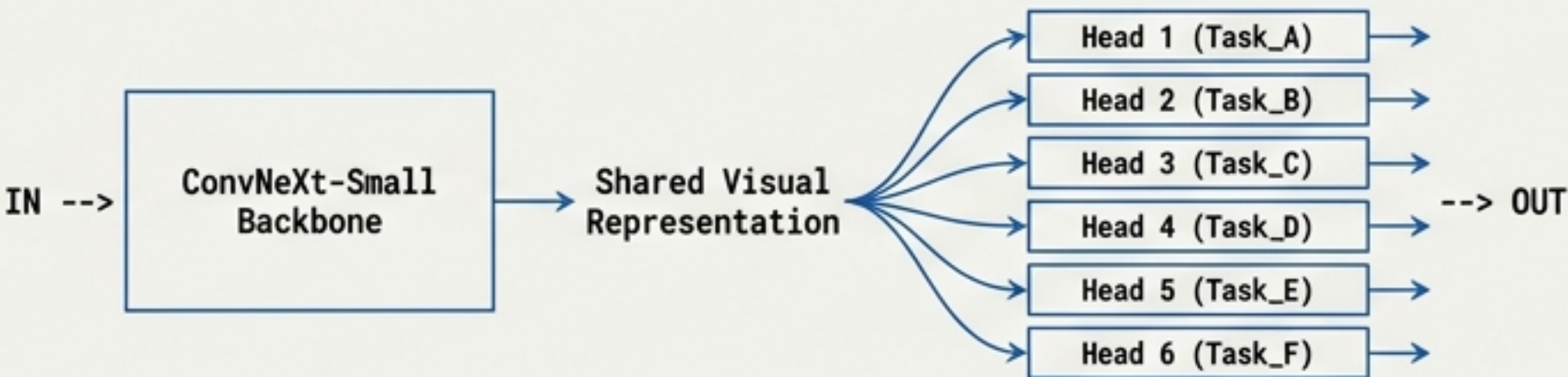
The 60-Second Elevator Pitch



The Architecture

Single Hard-Parameter Sharing Multi-Task CNN

A ConvNeXt-Small backbone extracts one shared visual representation, feeding 6 independent, lightweight linear heads.

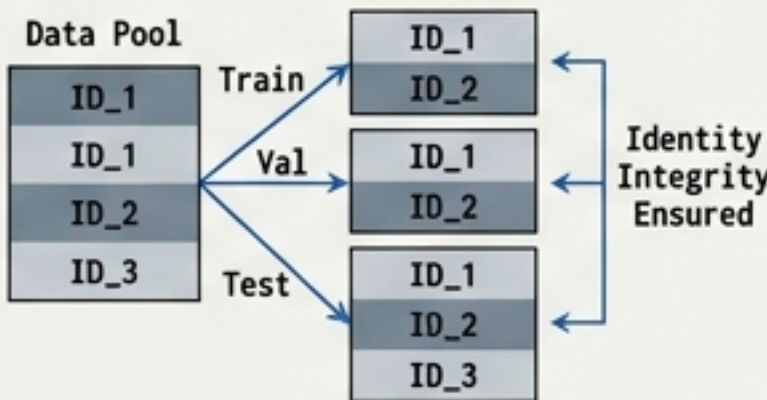


The Data Strategy

Leakage-Safe & Color-Safe

GroupShuffleSplit prevents Market-1501 identity leakage. Augmentations simulate CCTV quality but strictly avoid hue shifts that corrupt clothing labels.

Data Splitting (GroupShuffleSplit)



Safe Augmentations (CCTV Sim.)



Original Color Preserved

UNSAFE (Hue Shift)



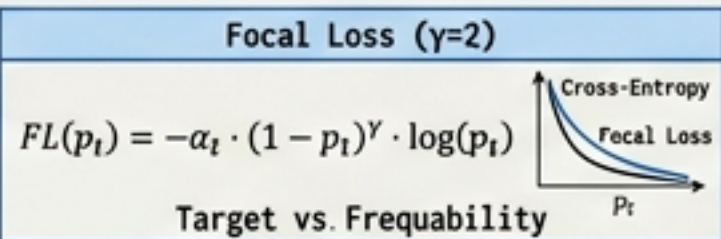
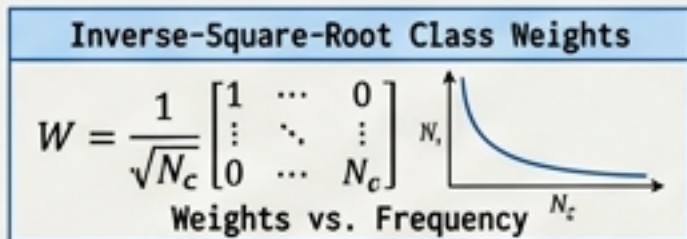
Clothing Labels Corrupted!



The Imbalance Solution

Tri-Axis Imbalance Stack

A synthesized attack using Inverse-Square-Root class weights, Focal Loss (gamma=2), and dynamic Homoscedastic Uncertainty weighting across tasks.



Focuses on Hard Examples

Dynamic Homoscedastic Uncertainty Weighting

$$L_{total} = \sum \frac{L_i}{2\sigma_i^2} + \log \sigma_i$$

Weights: $L_1, L_3, \dots, L_6$
 $L_1, L_2, \dots, L_6$
 $\sigma_1, \sigma_3, \dots, \sigma_6$

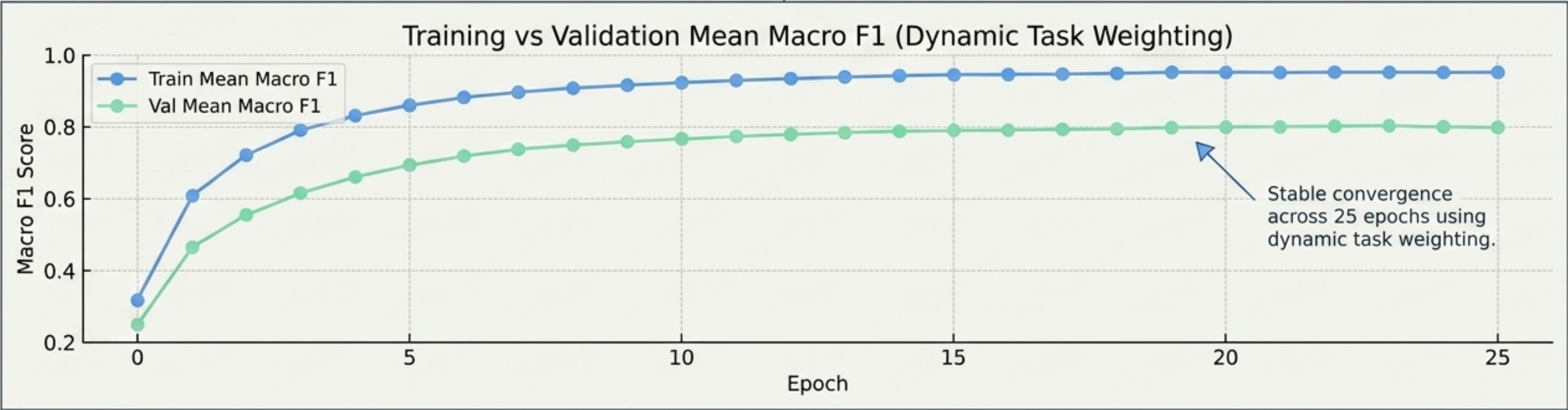
Balances Task Contribution

Final Results Dashboard

0.8216

Test Mean Macro-F1

Attribute	Accuracy	Macro-F1
Age	0.9897	0.9262
Headgear	0.9880	0.8756
Gender	0.9459	0.9449
Glasses	0.8335	0.7316
UBColor	0.8050	0.7534
LBColor	0.7909	0.6977

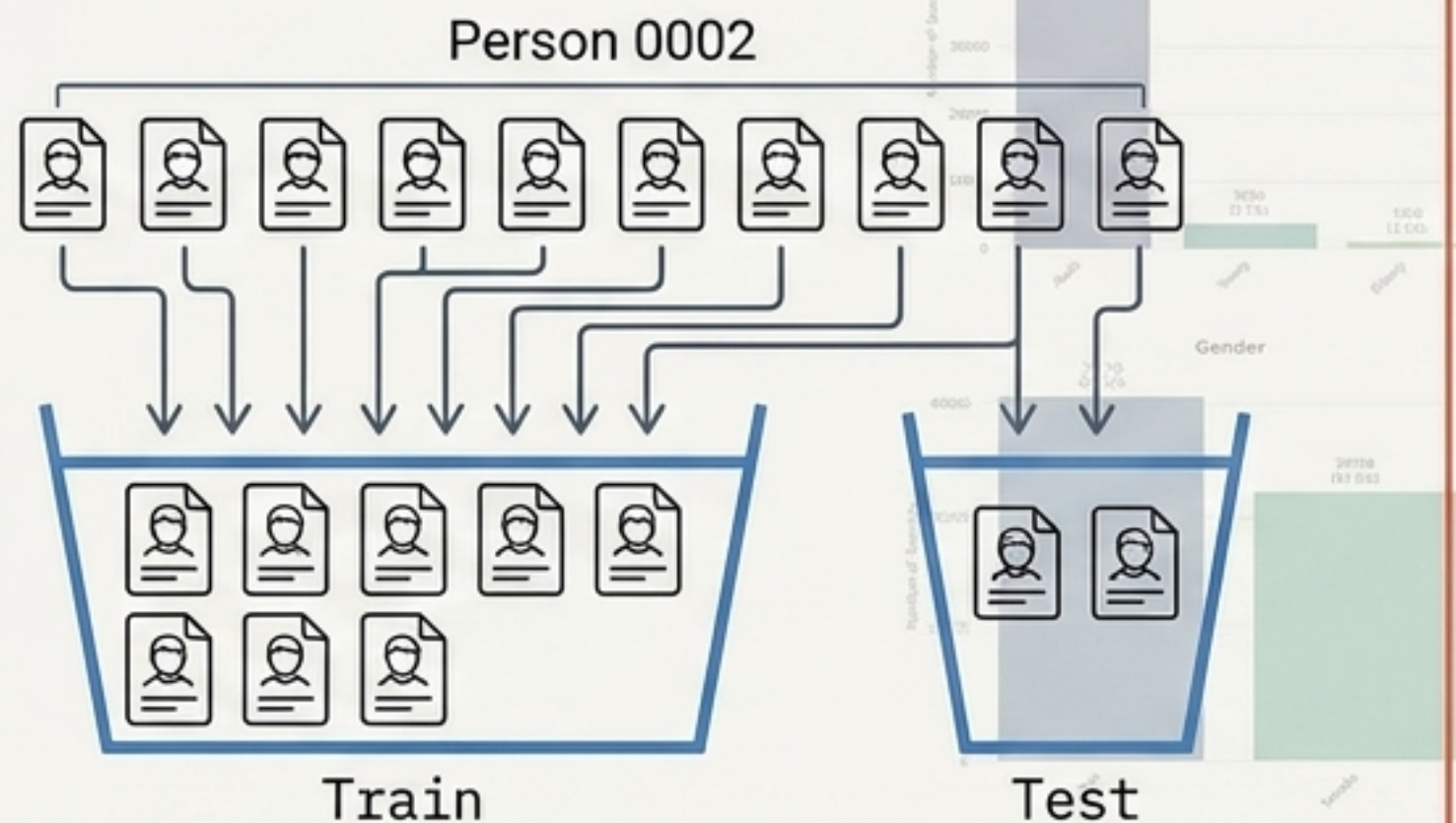


The Threat of Identity Leakage

The dataset has 70k images but fewer unique people. The filename 0002_c1s1 reveals Person ID 0002.

The Trap: DO NOT DO THIS

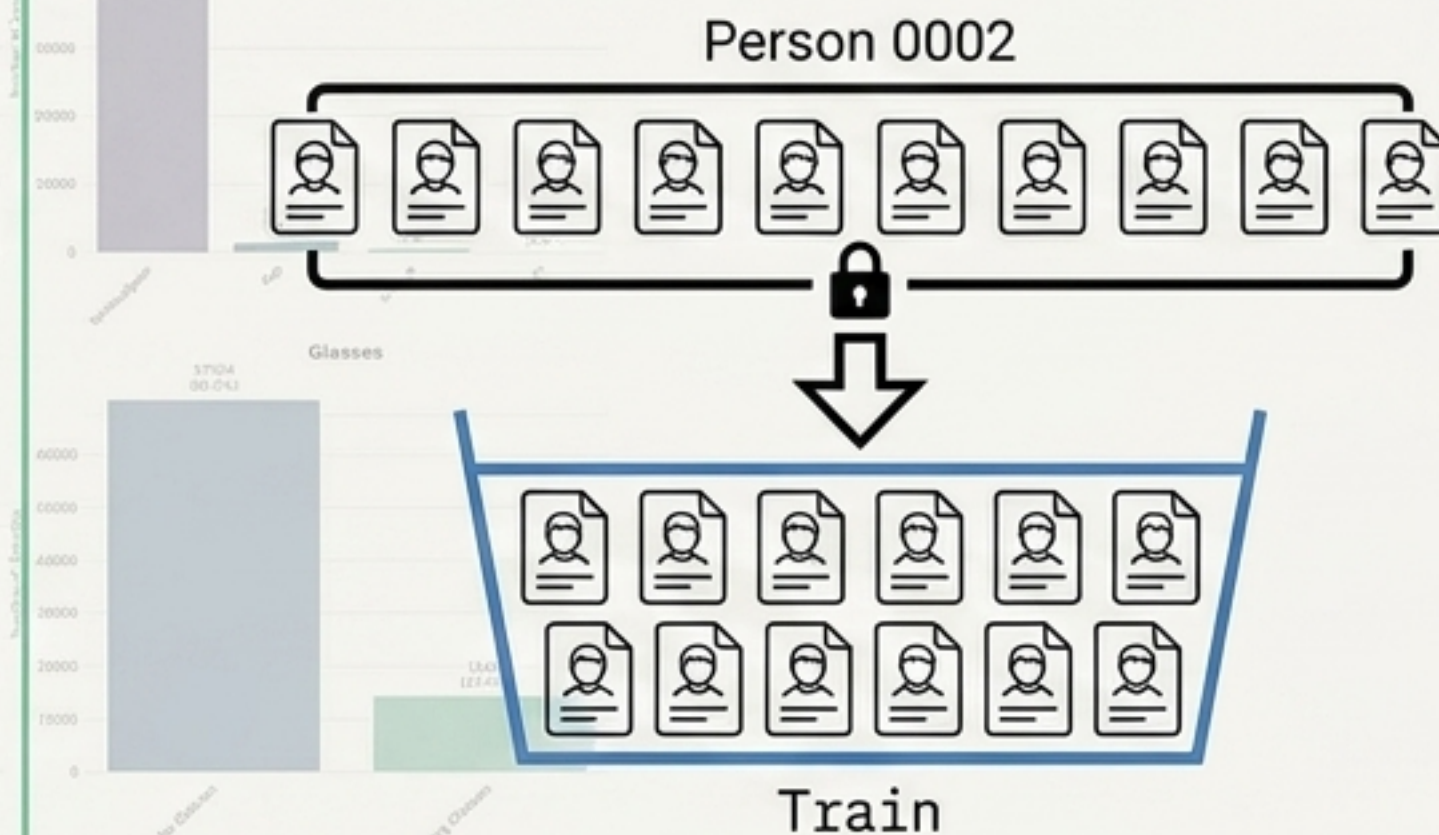
`train_test_split`



Result: Identity Leakage. The model lazily memorizes Person 0002's face instead of learning the universal visual features of a Red Shirt.

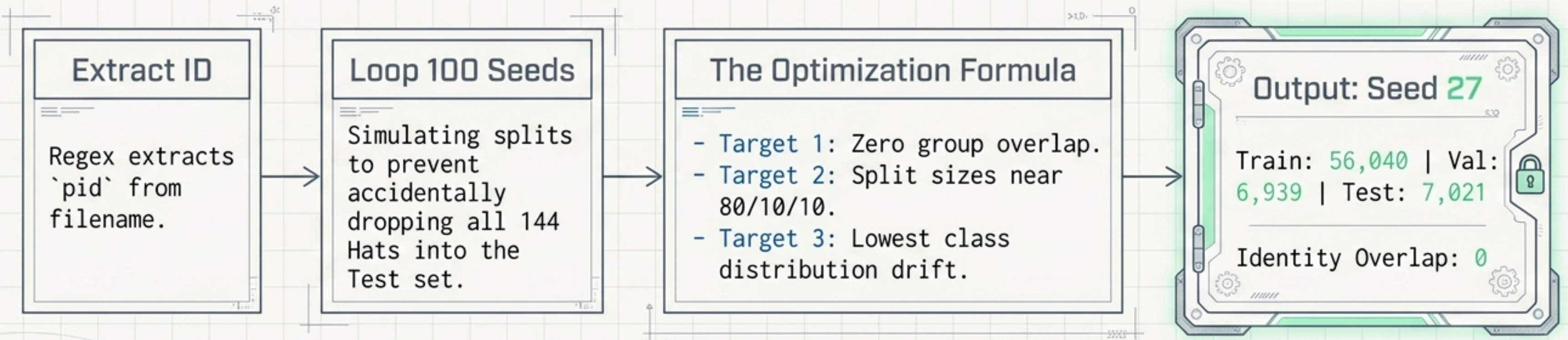
The Fix

`GroupShuffleSplit`



Result: The model is evaluated on 100% unseen strangers.

GroupShuffleSplit & The Seed Search



The Visual Defense: Preprocessing & Augmentation

Approved Augmentations (Domain Robustness)

A.Resize(256, 128)

Humans form a 2:1 vertical rectangle. Squashing to 224x224 destroys anatomical geometry.

**A.GaussNoise &
A.ImageCompression**

Simulates artifacts of cheap CCTV cameras.

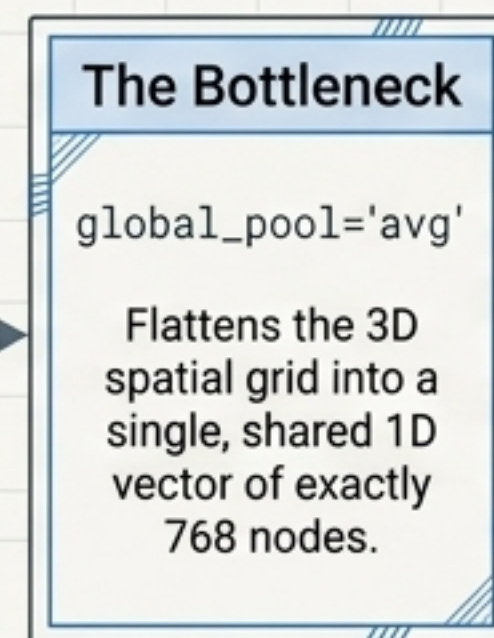
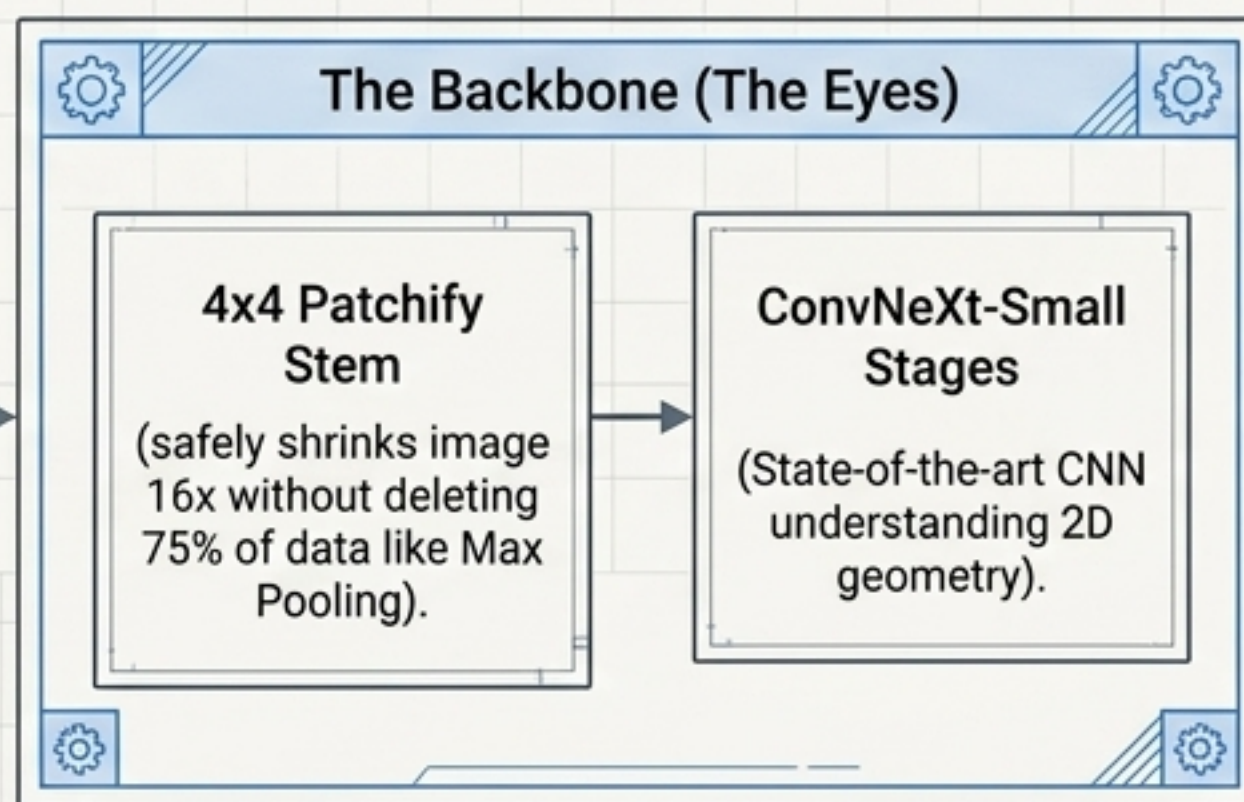
Banned Augmentations (Color-Safe Pipeline)



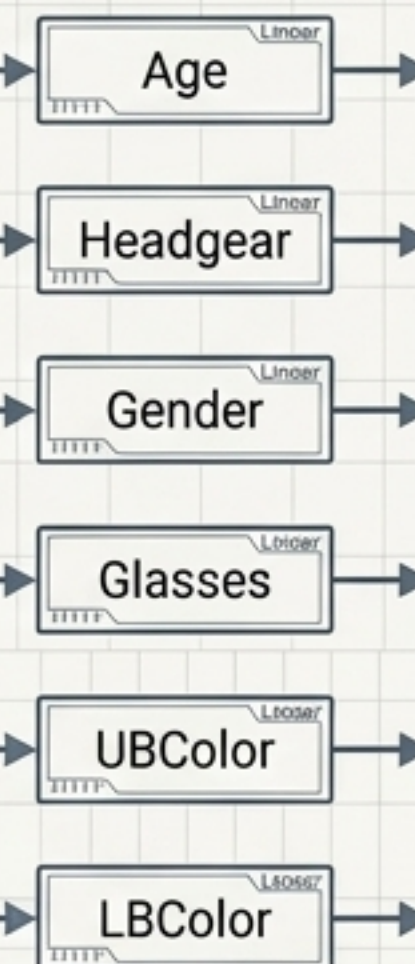
**NO ColorJitter.
NO Hue shifts.**

The Logic: UBColor and LBColor are target labels. Shifting the hue turns a red shirt green, but the CSV label still reads Red. Hue shifts poison the dataset.

Hard-Parameter Sharing Blueprint



The Heads (The Brains)



`num_classes=0` guarantees the original ImageNet head is removed.

The Tri-Axis Imbalance Attack

Extreme Imbalance: 67,515 NoHeadgear vs 144 Hats (469:1 ratio)

MODEL ACCURACY

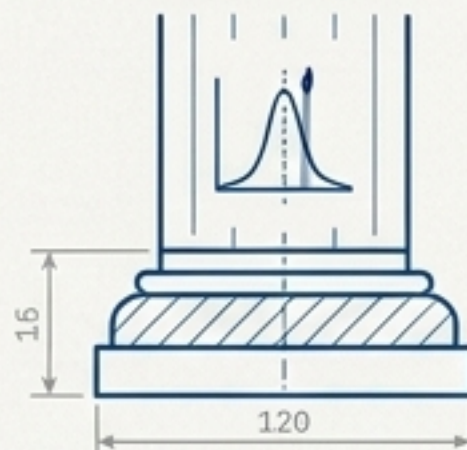
Axis 1: Within-Class Prior

Inverse-Square-Root
Class Weights.
(Attacking representation
volume).



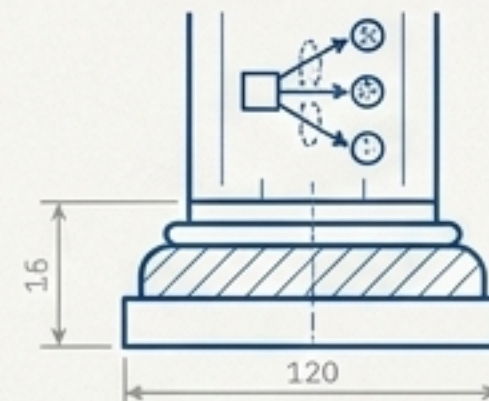
Axis 2: Within-Class Posterior

Focal Loss.
(Attacking prediction
confidence).



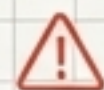
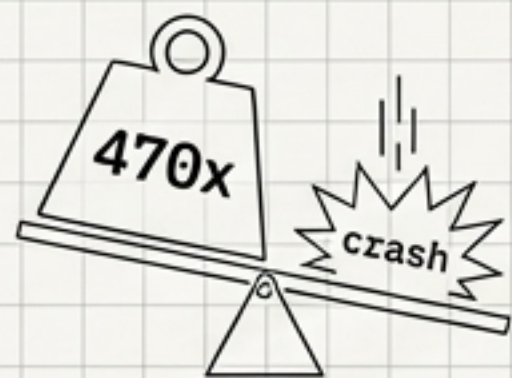
Axis 3: Across-Task

Homoscedastic
Uncertainty Weighting.
(Attacking negative transfer
between different attributes).



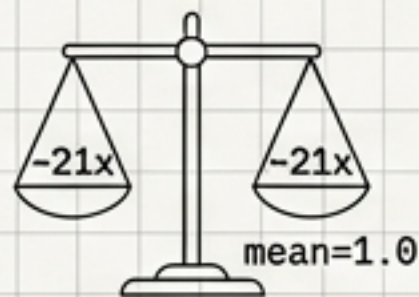
Axis 1 & 2: Weights & Focal Loss

Panel 1: Inverse-Sqrt Weights



The Trap:

Pure inverse frequency yields a 470x penalty weight for Hats. Gradients explode; model crashes.



$$Weight_{class} = \sqrt{\frac{Total_samples}{N_{class}}}$$

$$Normalized_Weight = \frac{Weight_{class}}{mean(Weight_{classes})}$$



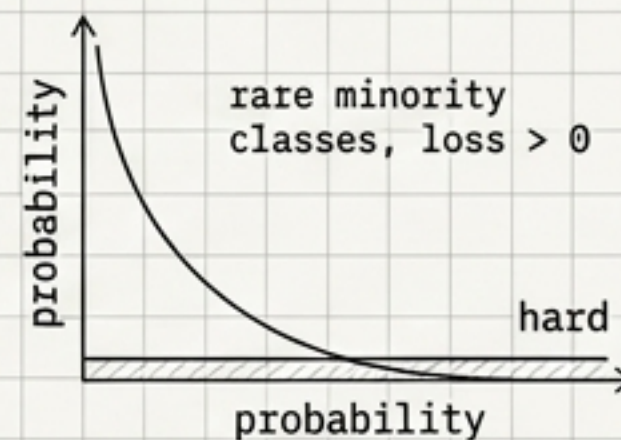
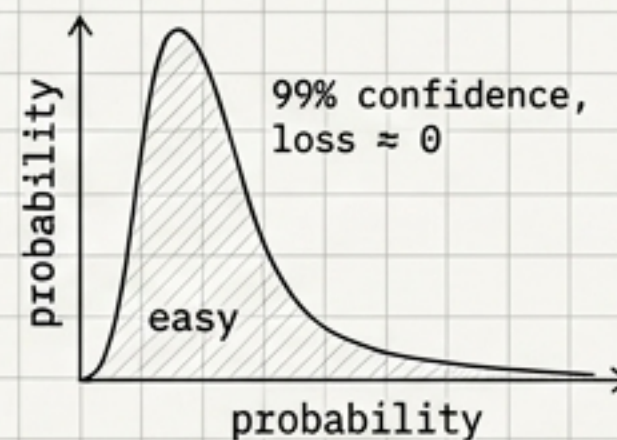
The Fix:

Square root softens the penalty to a safe ~21x multiplier. Normalization (mean=1.0) prevents weights from accidentally acting as massive learning-rate multipliers.

Panel 2: Focal Loss (gamma=2)

$$\text{CrossEntropy} * (1 - \text{probability})^2$$

Loss = $-(1 - p_t)^{\gamma} + \log(p_t)$
where p_t is model's estimated probability for the target class.

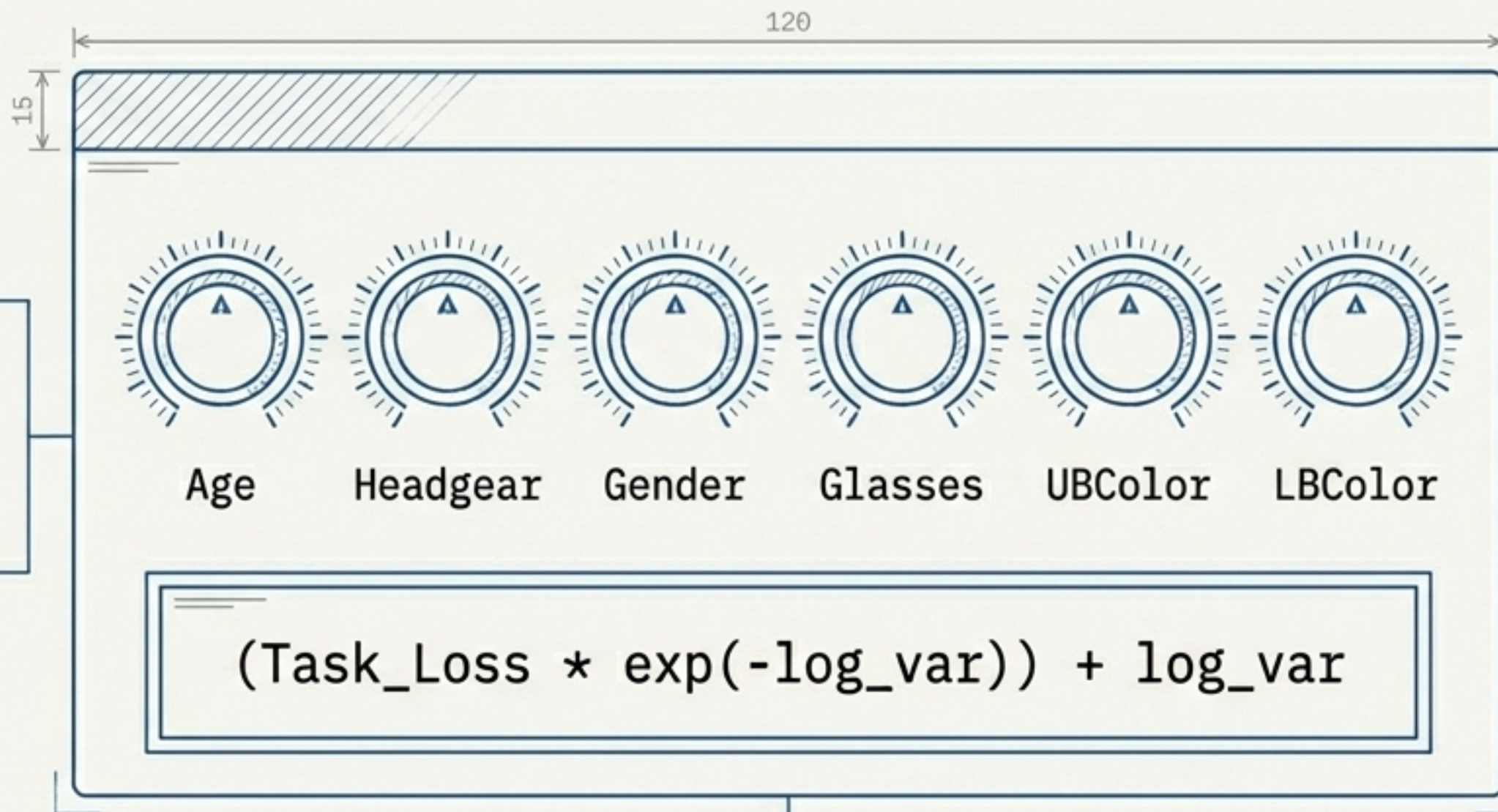


The Effect:

Forces the model to ignore easy 99%-confidence NoHeadgear guesses (loss drops near zero) and spend 100% of gradient updates learning the hard, rare minority classes.

Axis 3: Homoscedastic Uncertainty Weighting

Mechanism:
6 learnable
parameters
(log_vars).



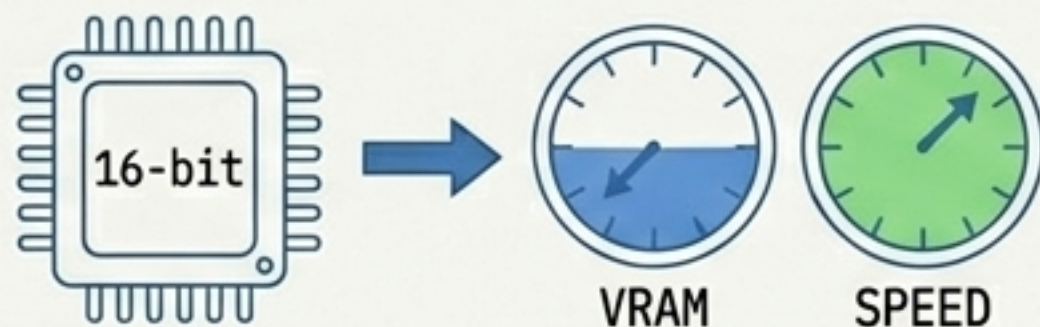
Explanation: The model acts as a DJ, dynamically turning down the gradient volume of noisy/fast-learning tasks to prevent Negative Transfer (where Age steals gradients from Clothing Color).

Proof: Why does total loss go negative?
As Task Loss nears zero, the optimizer pushes the log_var dials below zero to minimize the total equation.
Negative loss is proof of successful dynamic weighting!

The PyTorch Engine (AMP & Optimizer)

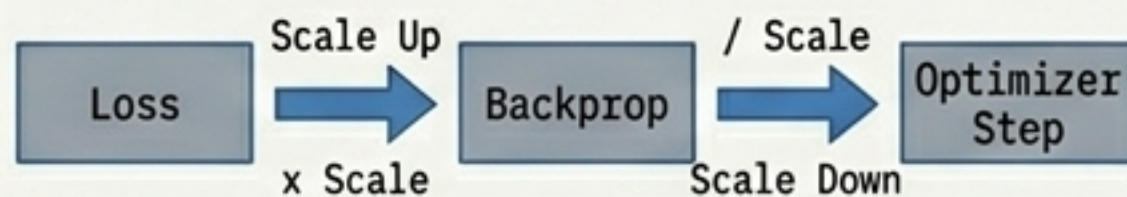
Automatic Mixed Precision (AMP)

Forces convolutions into 16-bit math (`torch.amp.autocast`).
Cuts VRAM in half, doubles speed.



GradScaler

Prevents 16-bit underflow (rounding down to 0.0) by scaling loss up before backprop and down before optimizer steps.

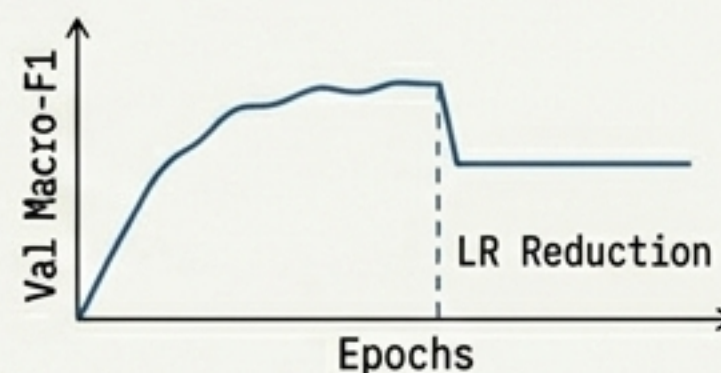


Optimizer

AdamW decouples weight decay for better transfer-learning generalization.

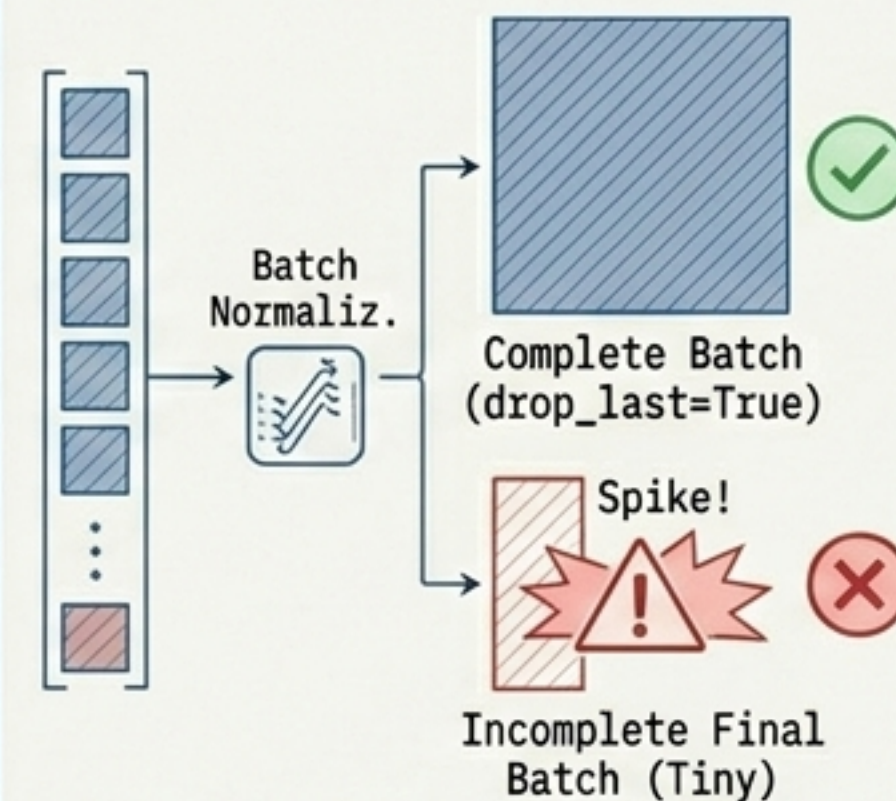


`ReduceLROnPlateau` monitored on Val Macro-F1.



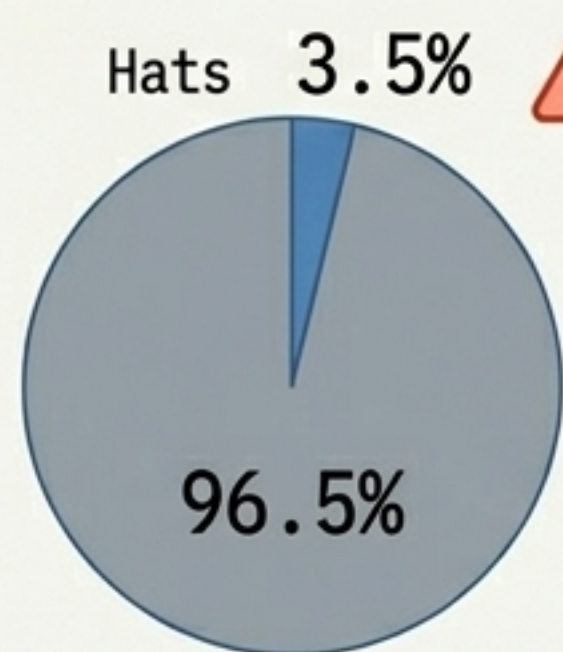
Stability

`drop_last=True`: Prevents mathematical spikes in Batch Normalization caused by tiny, incomplete final batches.



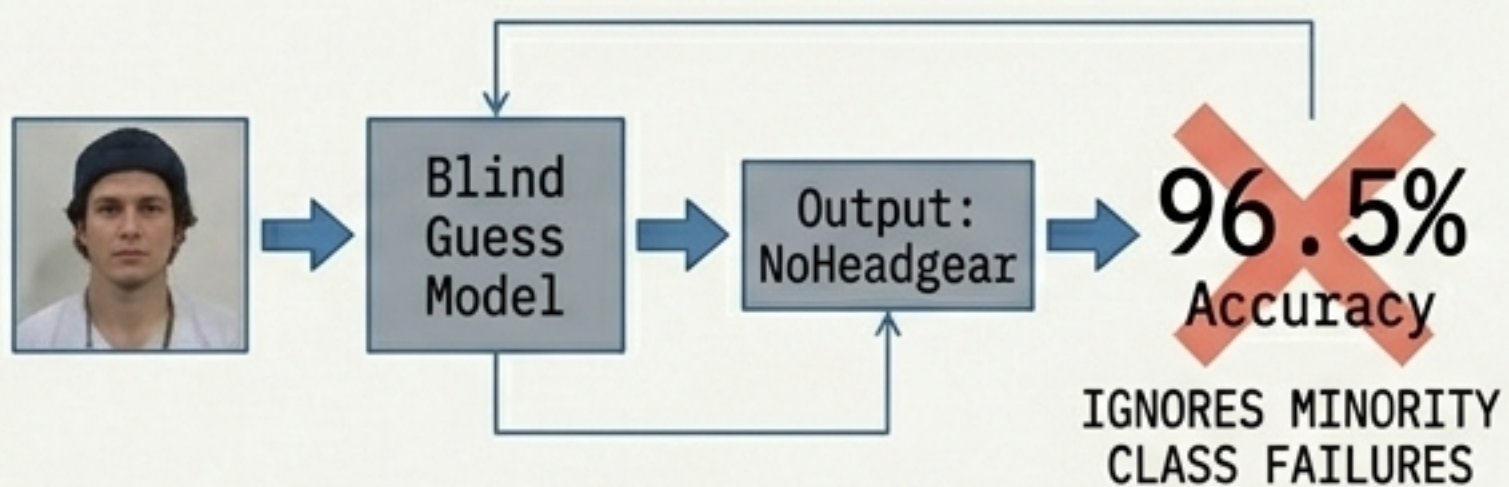
Evaluation Metrics: The Macro-F1 Truth

The Accuracy Lie



! If 96% of subjects have NoHeadgear, a completely broken model that guesses NoHeadgear blindly on every image will achieve 96.5% Accuracy.

Accuracy hides minority failure.



The Macro-F1 Truth

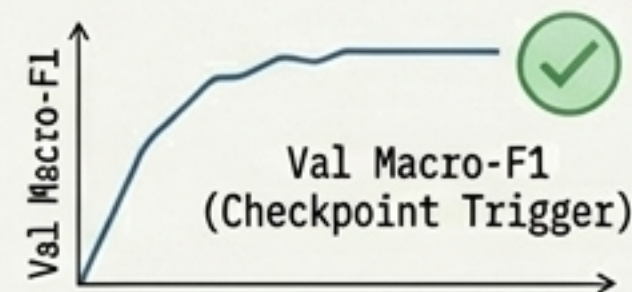


Calculates F1 for the majority class and minority class separately, then averages them equally.

Macro-F1

Macro-F1 is the only metric that forces the model to respect rare classes like Hats.

Checkpointing is strictly monitored on `val_mean_macro_f1`.



Failure Mode Analysis (Domain Shift)

DOMAIN SHIFT: IMAGE COMPARISON



DOMAIN SHIFT: EXPLANATION

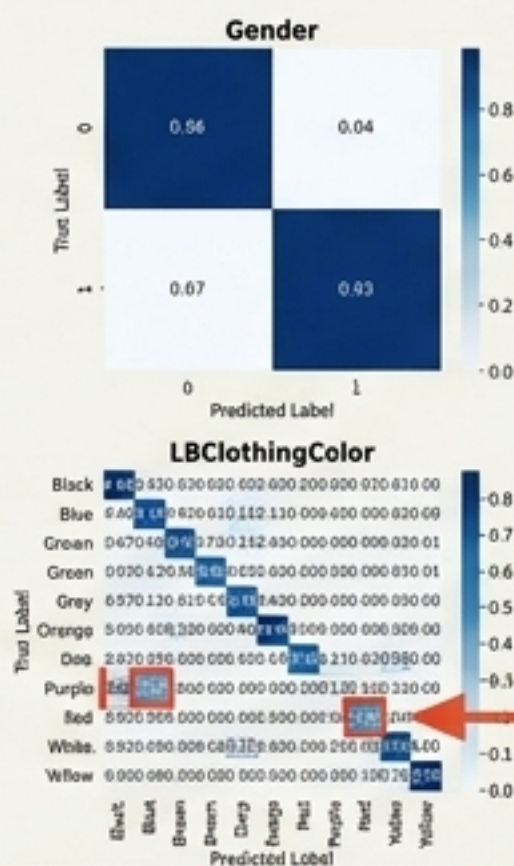
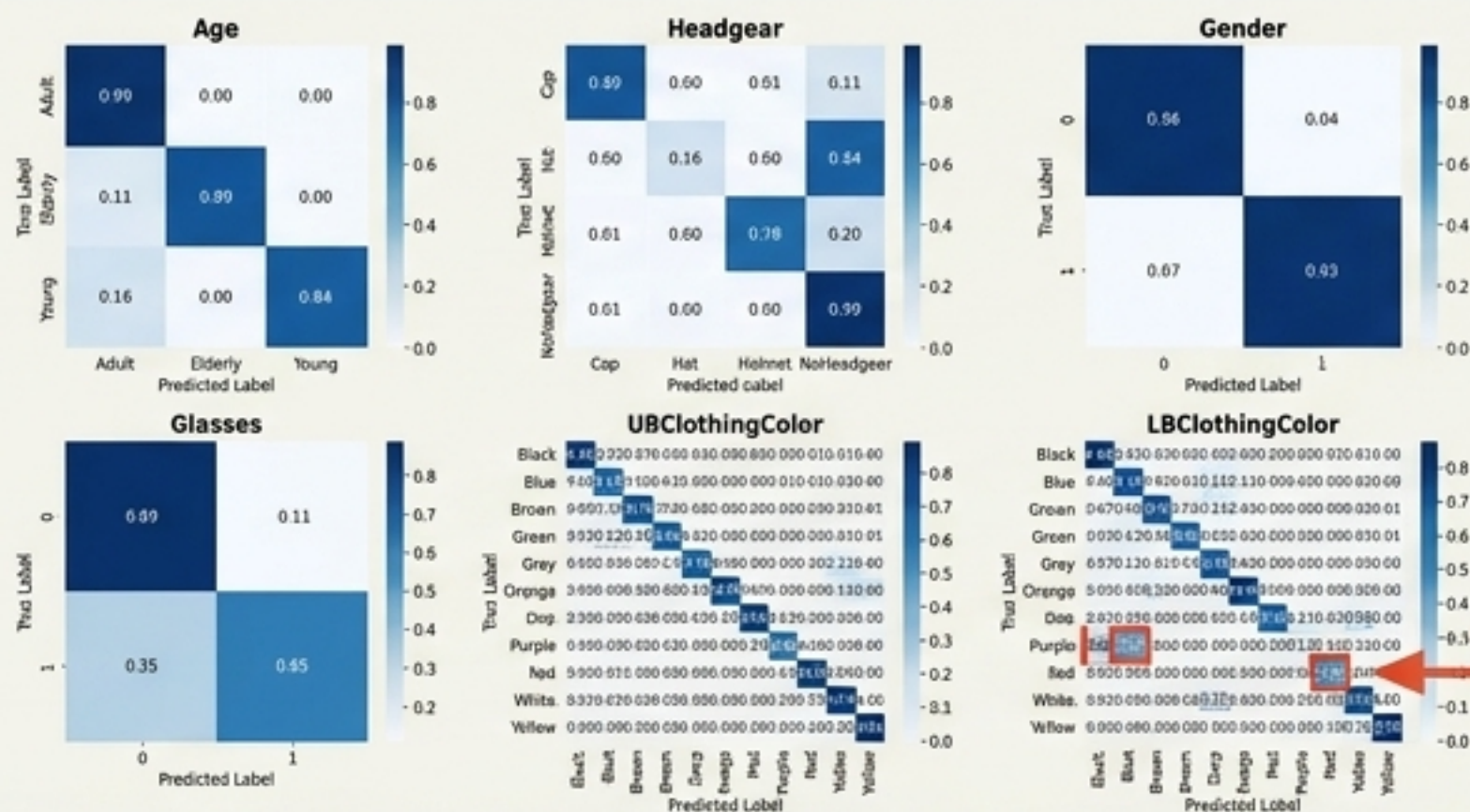
The model was trained on low-resolution, blurry, top-down CCTV cameras. External testing uses HD, eye-level photography.

The visual gap forces the model into low-confidence, default majority-class guesses.

ILLUMINATION VARIANCE: EXPLANATION

Shadows change raw pixel data (e.g., a black shirt in bright sunlight reflects white/blue pixels).

The confusion matrix reveals Purple (98 samples) is misclassified as Blue 50% of the time.



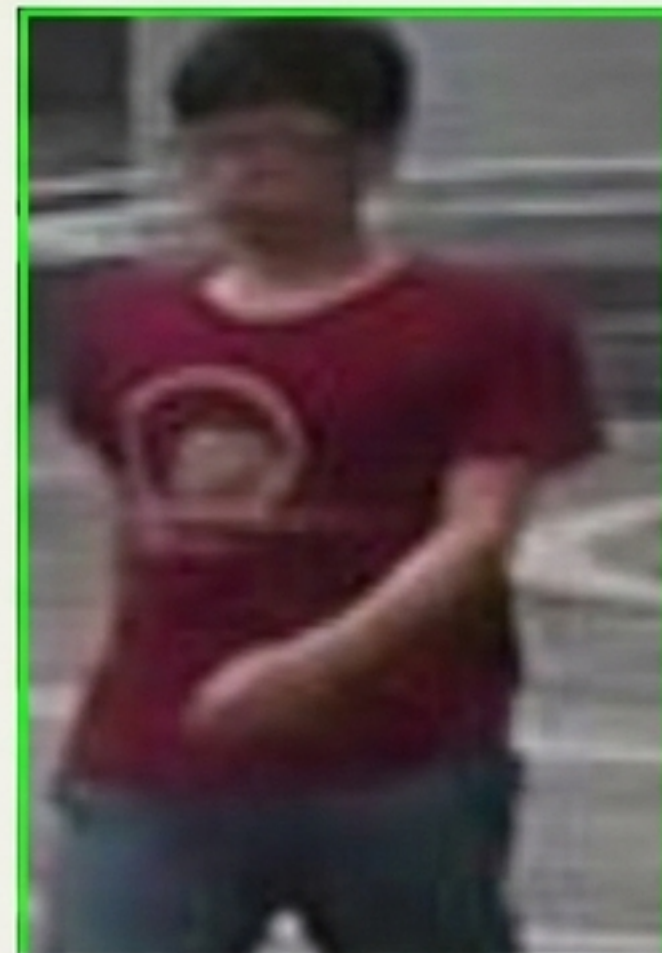
The Why Not? Grilling Matrix

Alternative Model	The Candidate's Constraint	The Verdict
YOLO	An Object Detector meant to draw boxes in crowded rooms. Our dataset is already cropped.	Architecturally incorrect. ⚠️
Vision Transformer (ViT)	Data-hungry; lacks spatial inductive bias.	CNNs are far more efficient for 70k static images. ⚠️
LoRA	Freezes 8B-parameter foundation models to fit on small GPUs.	ConvNeXt-S is only 50M parameters and fits easily on a T4. Full fine-tuning yields maximum accuracy. ✅
RNN	For time-series/video sequences.	Irrelevant for static 2D JPEGs. ⚠️
6 Separate Models	Wastes compute, violates assignment rules, ignores shared facial/clothing features.	Inefficient and misses cross-task synergies. ⚠️

The Master Pipeline Summary

1. Spotted Market-1501 identity leakage in filenames.
2. Isolated identities with a 100-seed GroupShuffleSplit.
3. Engineered color-safe augmentations (protecting hue).
4. Built a single hard-parameter sharing ConvNeXt backbone.
5. Handled 469:1 imbalances via Inverse-Sqrt + Focal Loss.
6. Dynamically weighted 6 tasks via Homoscedastic Uncertainty.

You typed the code. You own the logic. You are ready.



Age: Adult
Headgear: NoHeadgear
Gender: 0
Glasses: 0
UBClothingColor: Red
LBClothingColor: Blue